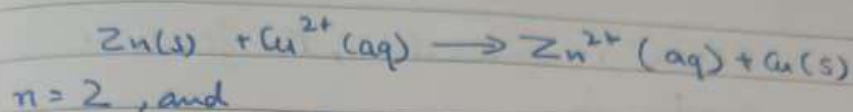


① For cell reaction



$$Q = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = \frac{0.10}{x}$$

Nernst equation at 298 K

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{2} \log Q$$

$$1.07 = 1.10 - \frac{0.0591}{2} \log \left(\frac{0.10}{x} \right)$$

Thus,

$$\log \left(\frac{0.10}{x} \right) = \frac{2(1.10 - 1.07)}{0.0591} = 1.015$$

$$\frac{0.10}{x} = 10^{1.015} = 10.35$$

$$x = \underline{\underline{9.66 \times 10^{-3} \text{ M}}}$$

② first order reaction

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

75% completion, $a-x = 0.25a$,

$$k = \frac{2.303}{20} \log \frac{a}{0.25a}$$

$$= \frac{2.303}{20} \log 4$$

$$= 6.93 \times 10^{-2} \text{ min}^{-1}$$

93.75% completion, the fraction remaining is

$$1 - 0.9375 = 0.0625 = \frac{1}{16}$$

Therefore,

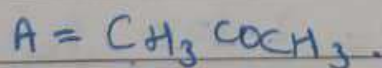
$$t = \frac{2.303}{k} \log 16$$

$$= \frac{2.303}{6.93 \times 10^{-2}} \log 16$$

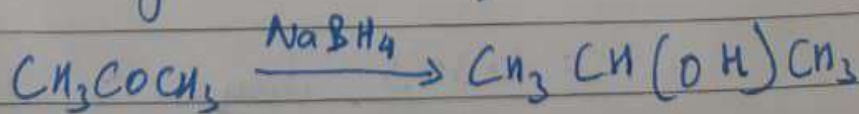
$$k = 6.93 \times 10^{-2} \text{ min}^{-1}$$

$$t_{93.75\%} = 40 \text{ min}$$

- ③ A positive 2,4 DNP test shows that A is a carbonyl compound. A negative Tollens' test rules out an aldehyde. Thus A is the ketone propanone.



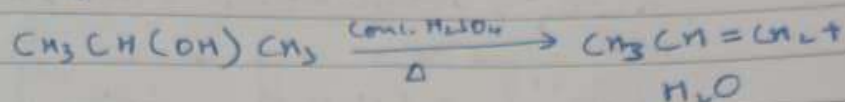
Reduction of propanone gives propan-2-ol:



Therefore,



On dehydration



A = propanone, B = propan-2-ol.

- (4) In both complexes cobalt is in the +3 oxidation states:



Complex	Crystal-field arrangement	Unpaired electron
$[\text{CoF}_6]^{3-}$	$t_{2g}^4 e_g^2$	4
$[\text{Co}(\text{NH}_3)_6]^{3+}$	$t_{2g}^6 e_g^0$	0

Hybridization	Magnetism
$sp^3 d^2$	Paramagnetic
$d^2 sp^3$	Diamagnetic

$[\text{CoF}_6]^{3-}$ is high spin & paramagnetic

$[\text{Co}(\text{NH}_3)_6]^{3+}$ is low spin & diamagnetic.

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⑤ Initial moles before mixing are.

$$n(\text{Ba}^{2+}) = 0.100 \times 0.020 = 2.00 \times 10^{-3} \text{ mol},$$

$$n(\text{SO}_4^{2-}) = 0.100 \times 0.010 = 1.00 \times 10^{-3} \text{ mol},$$

total volume after mixing is 0.200L. Thus before.

$$[\text{Ba}^{2+}] = 0.010 \text{ M}, \quad [\text{SO}_4^{2-}] = 0.0050 \text{ M}$$

Ionic product is.

$$Q_{sp} = (0.010)(0.0050) = 5.0 \times 10^{-5}$$

$Q_{sp} \gg K_{sp}$, BaSO_4 precipitates.

$$[\text{Ba}^{2+}] \approx \frac{2.00 \times 10^{-3} - 1.00 \times 10^{-3}}{0.200} = 0.0050 \text{ M}$$

$$K_{sp} = [\text{Ba}^{2+}][\text{SO}_4^{2-}]$$

$$[\text{SO}_4^{2-}] = \frac{1.1 \times 10^{-10}}{0.0050} = 2.2 \times 10^{-8} \text{ M}$$

Moles of residual sulfate =

$$(2.2 \times 10^{-8})(0.200) = 4.4 \times 10^{-9} \text{ mol.}$$

Hence

$$n(\text{BaSO}_4) \approx 1.00 \times 10^{-3} - 4.4 \times 10^{-9} = 9.9996 \times 10^{-4} \text{ mol}$$

$$\therefore [\text{SO}_4^{2-}] = 2.2 \times 10^{-8} \text{ M}$$
$$\approx 0.233 \text{ g}$$